

Sample End Semester Question Paper - 2025

Subject: Circuit Theory | Code: EE201 | Semester: 2

Branch: EEE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks: 100

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. What is the time constant of an RL circuit?
2. Define quality factor.
3. What is a node?
4. State the superposition theorem.
5. What is a phasor?
6. State Kirchhoff's current law.
7. What is a mesh?
8. Define impedance.
9. What is resonance?
10. State Thevenin's theorem.

PART B - Long questions (6 x 13 = 78 Marks)

1. Explain nodal and mesh analysis. Solve a given circuit.
2. Explain series and parallel RLC circuits. Derive impedance and admittance.
3. Explain Thevenin's, Norton's and superposition theorems with examples.
4. Explain transient analysis of first-order RL and RC circuits.
5. Explain resonance and quality factor in AC circuits.
6. Explain maximum power transfer theorem for AC circuits.